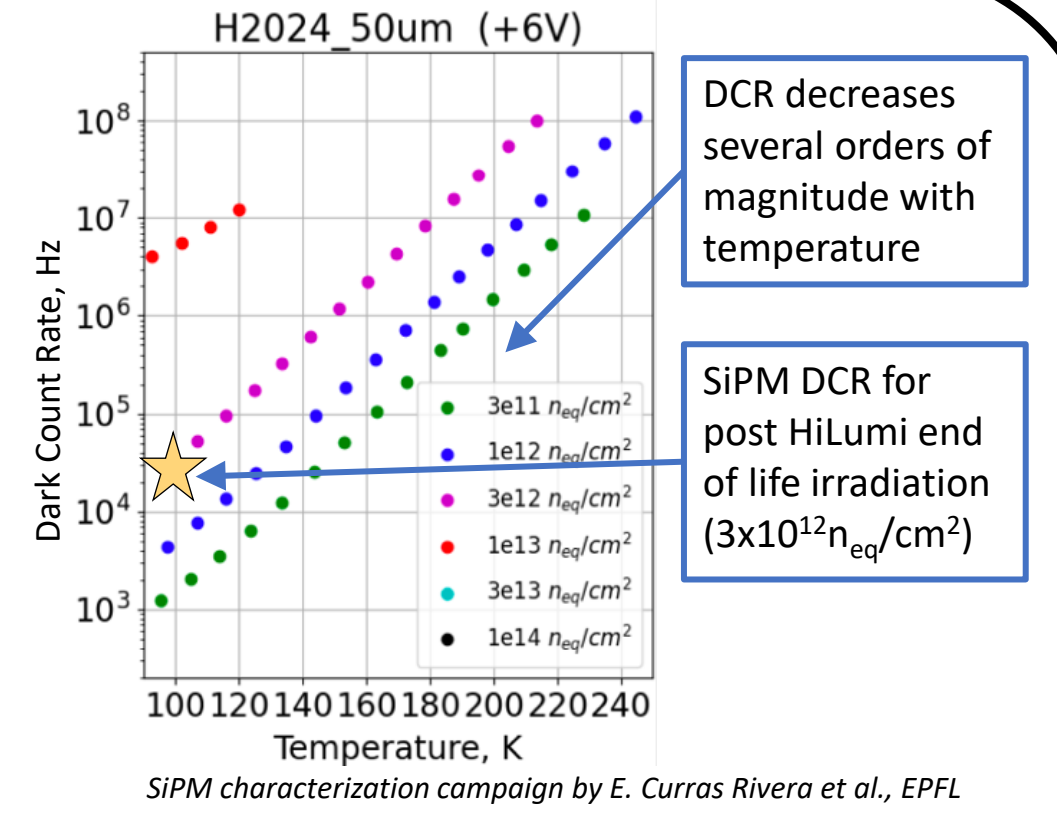
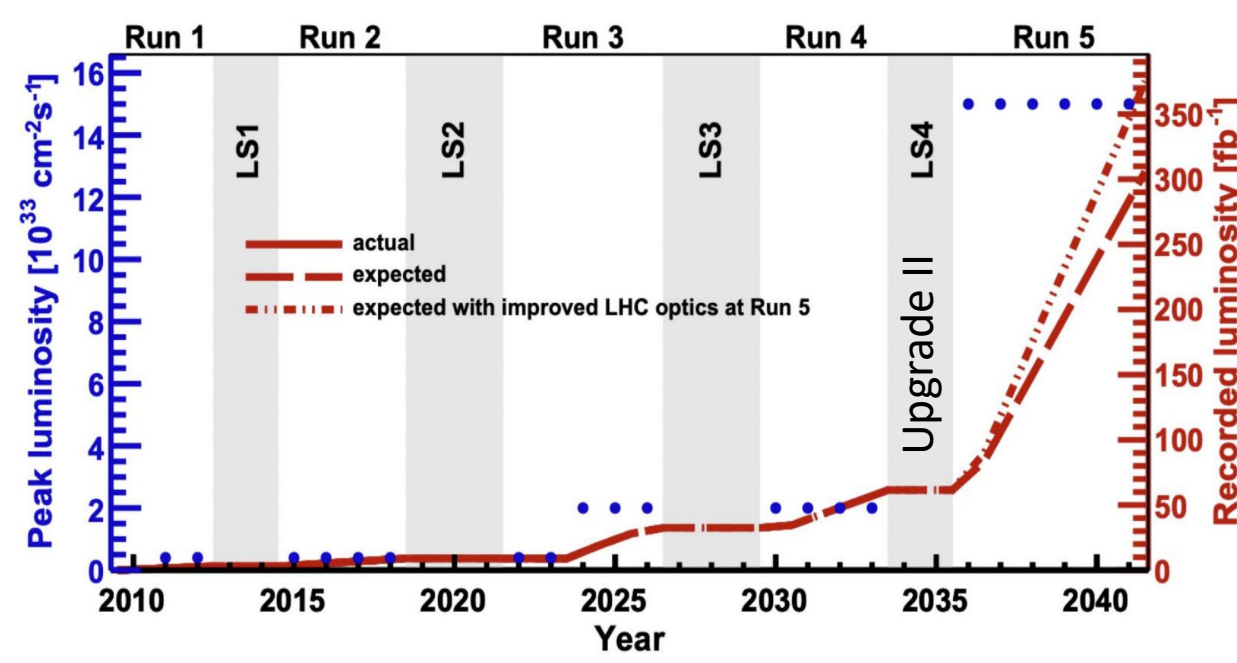
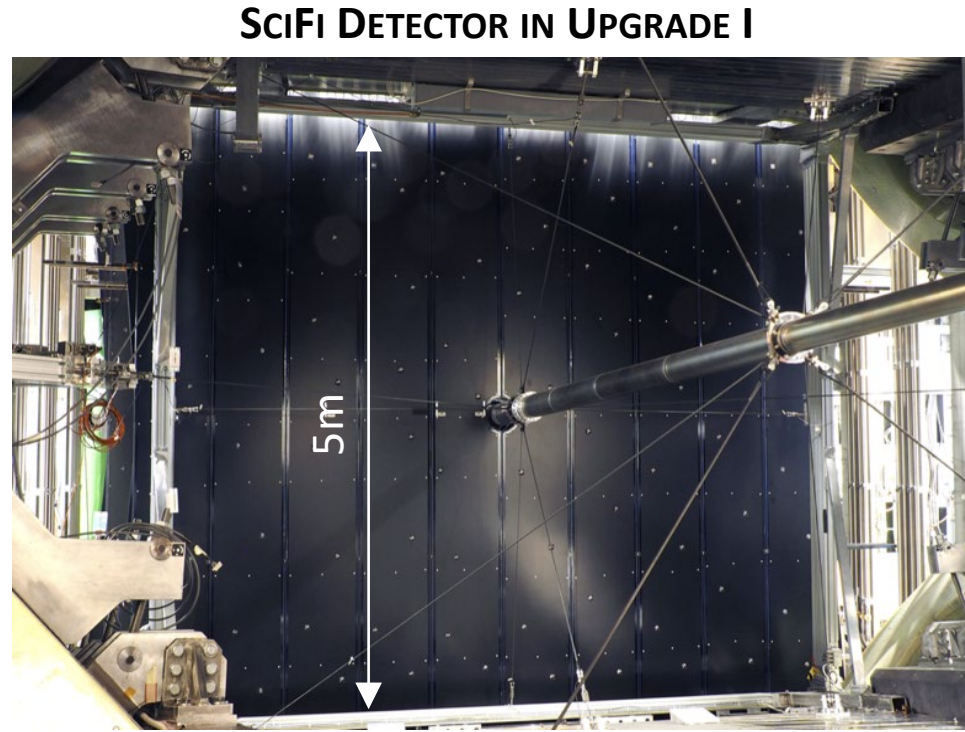
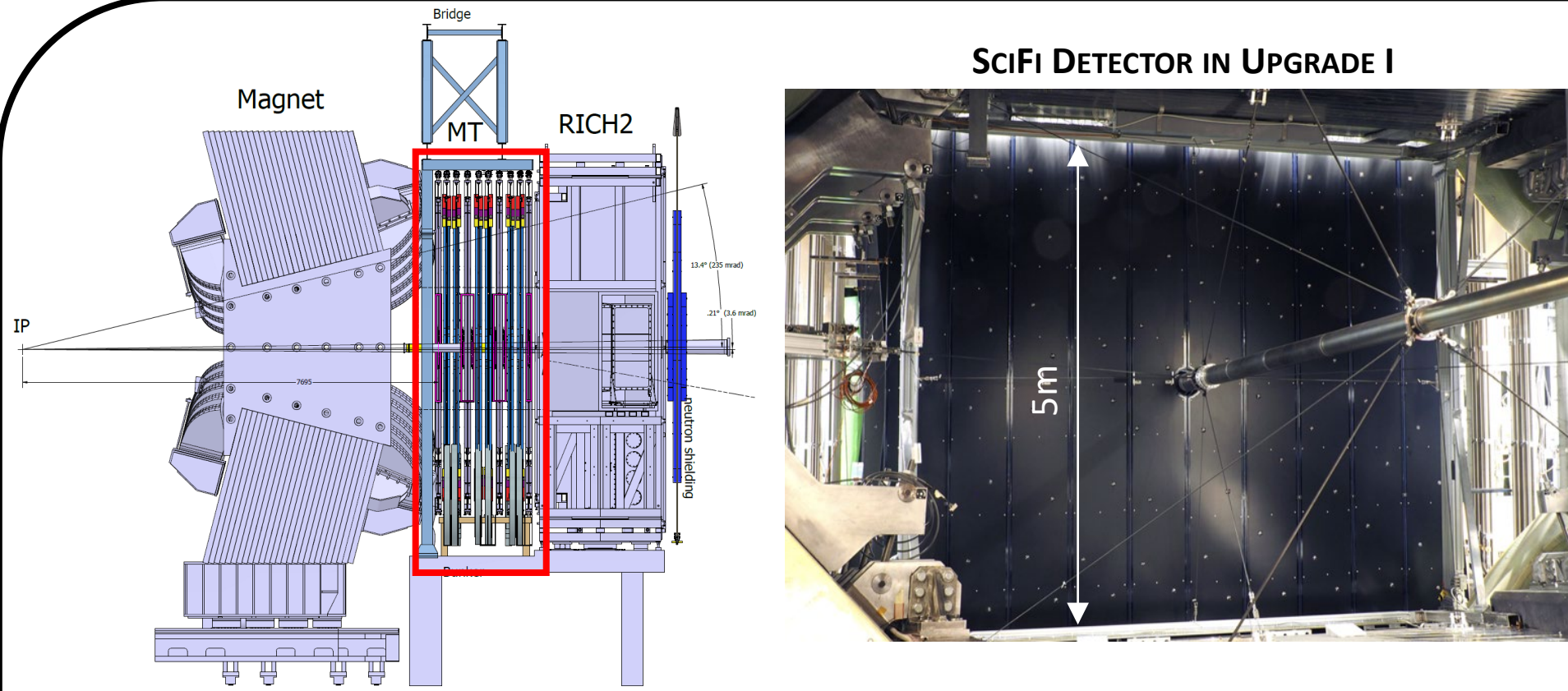


MIGHTY-SCI FI CRYO DEMONSTRATOR FOR LHCb UPGRADE II

Gauri Napoletano (EPFL) on behalf of Mighty-Tracker Collaboration

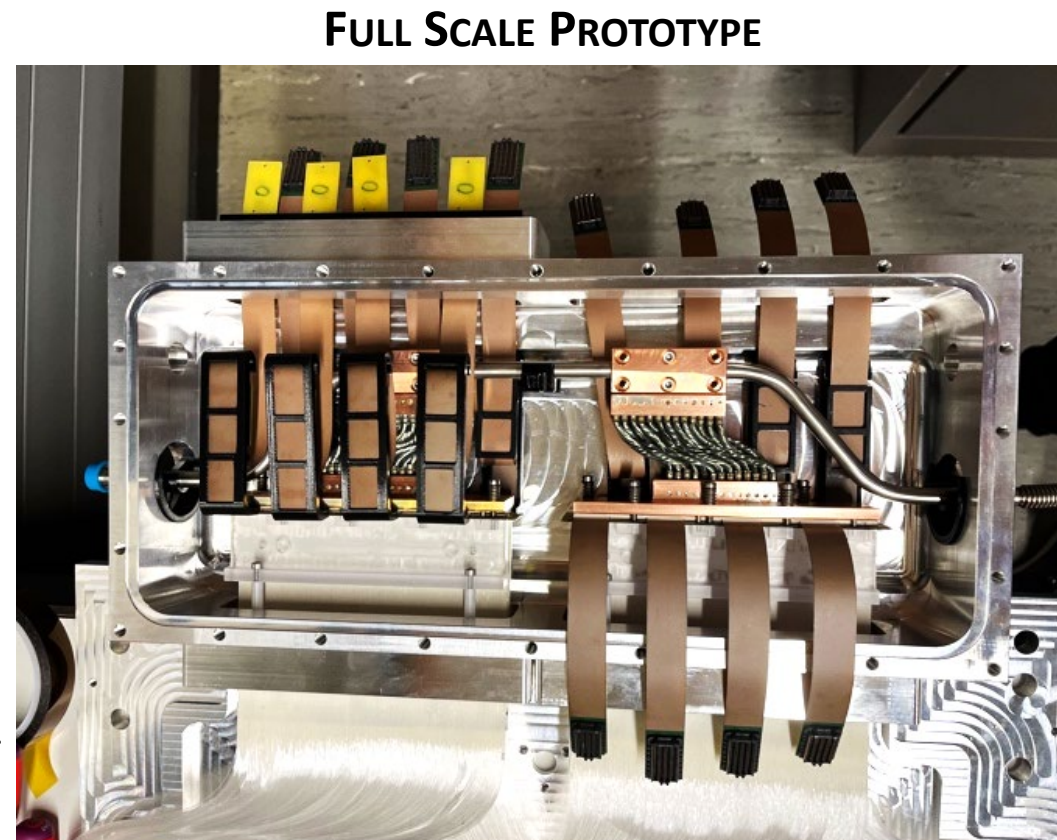
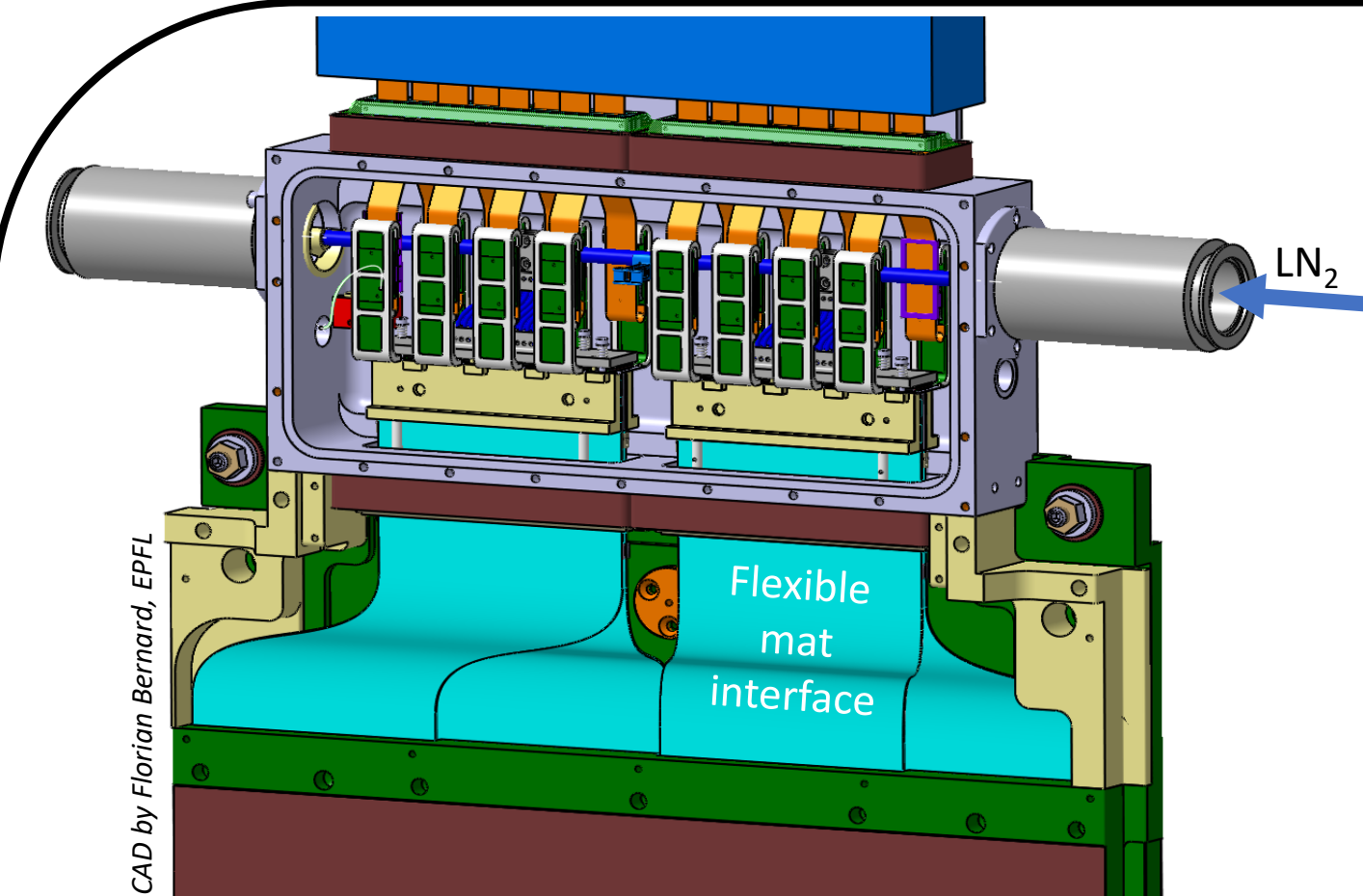
INTRODUCTION



- 250 μm scintillating fibres readout by multichannel SiPMs arrays
- Better than 100 μm spatial resolution
- 12 layers ($6 \times 5 \text{ m}^2$) for a total active area of 340 m^2
- SiPMs operated at -40°C in Upgrade I

- Due to radiation the SiPM noise is dramatically increased, expected neutron fluence is $3 \times 10^{12} n_{\text{eq}}/\text{cm}^2$ for Upgrade II
- Cooling to $<100 \text{ K}$ necessary to keep the DCR acceptably low afterwards
- Two demonstrators were built and their thermal performance was evaluated

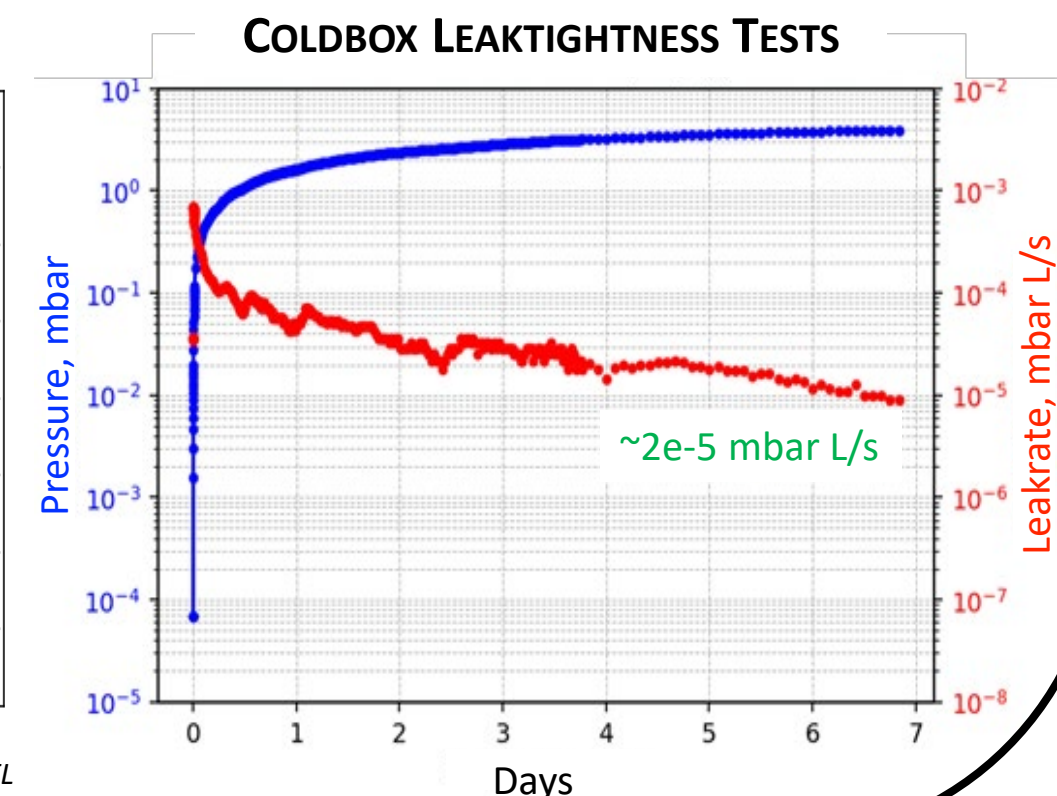
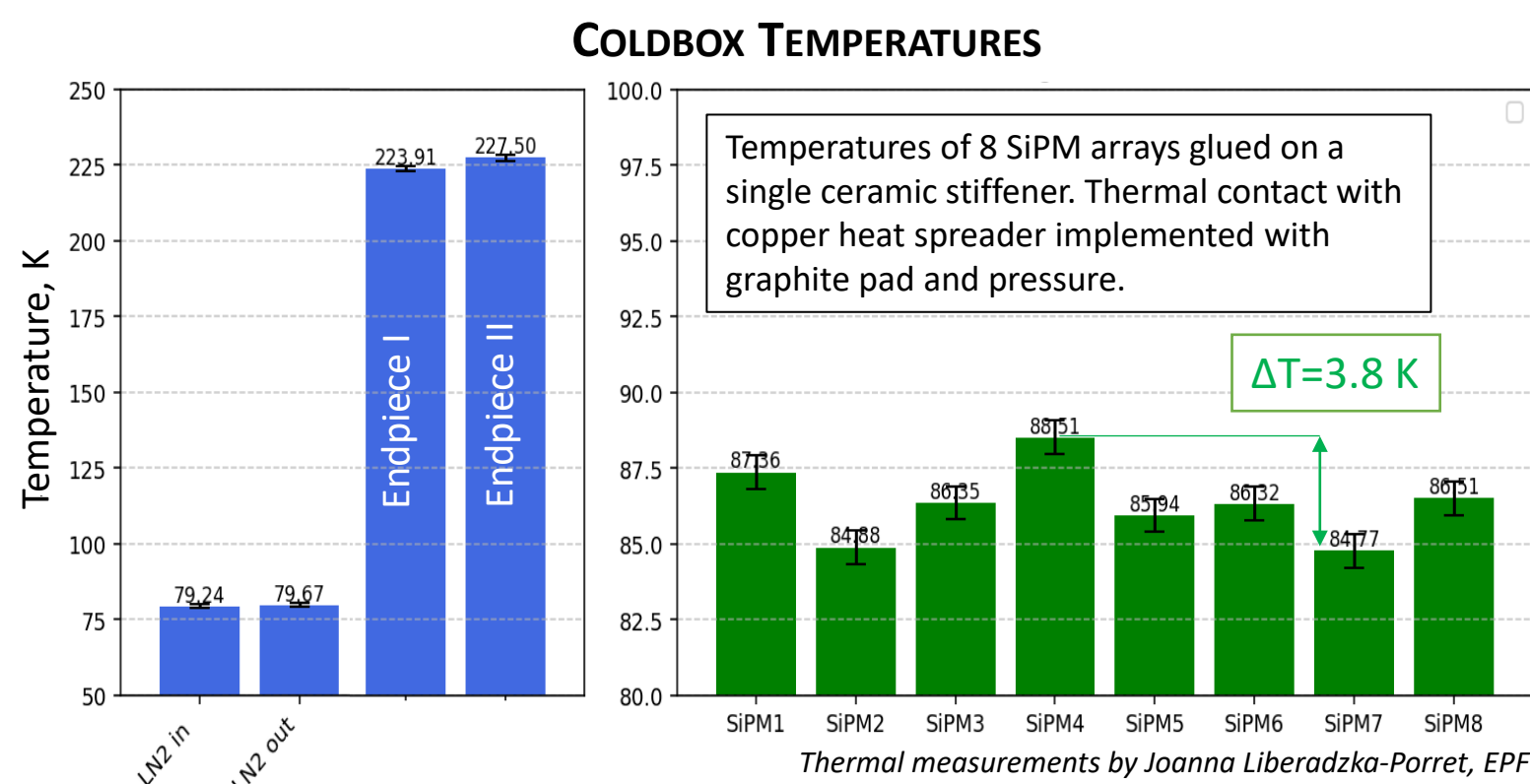
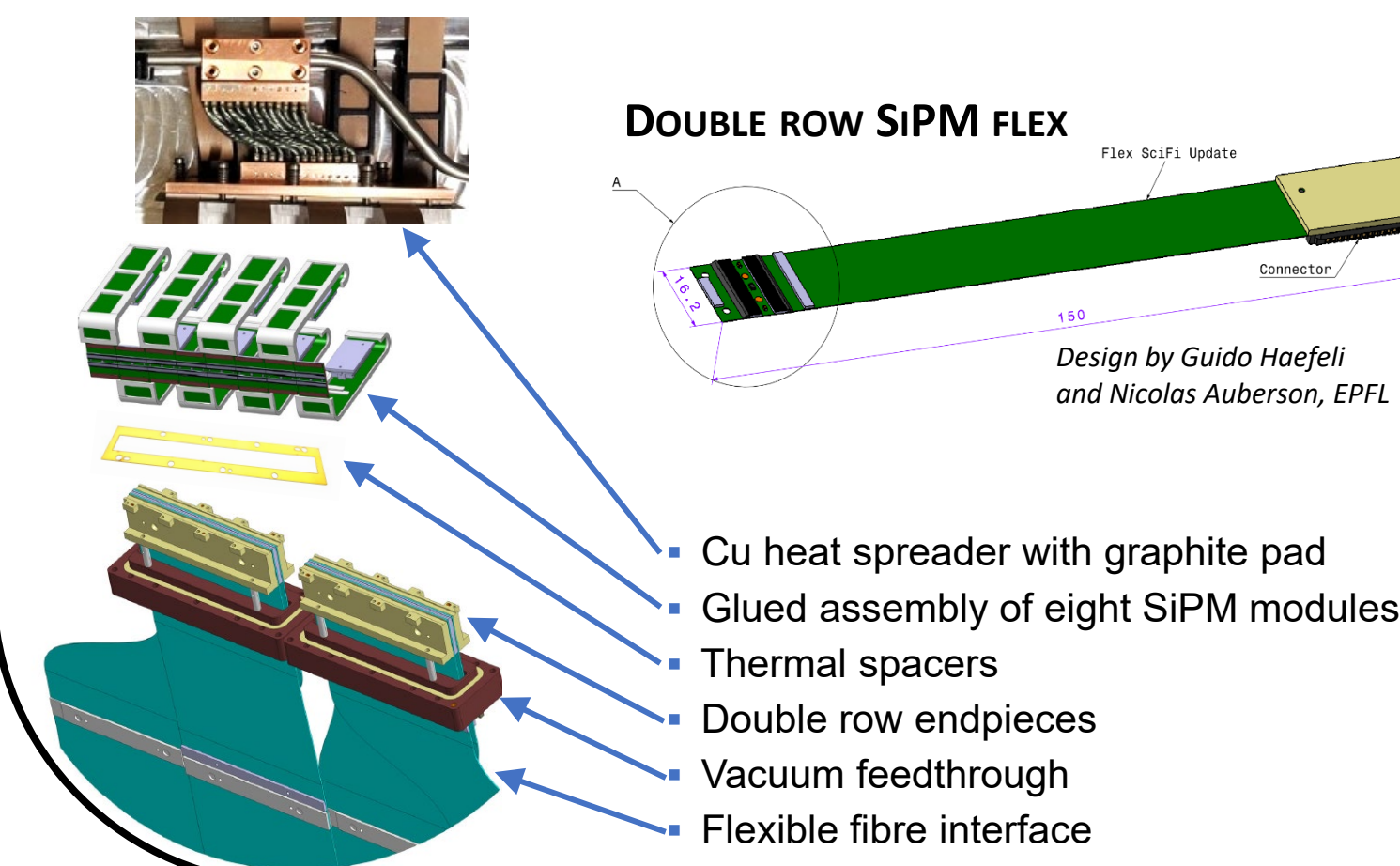
FLEXIBLE FIBRE INTERFACE



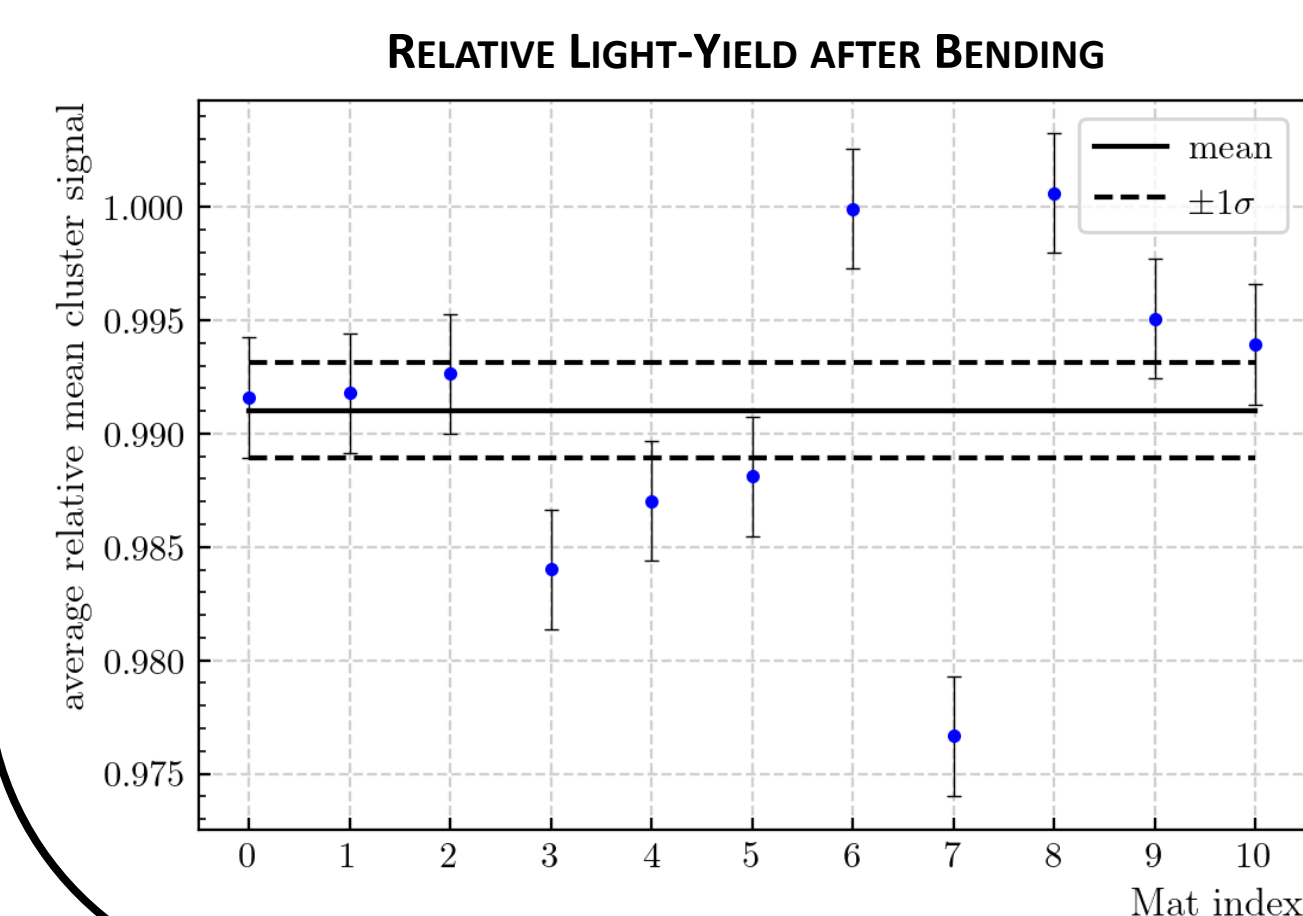
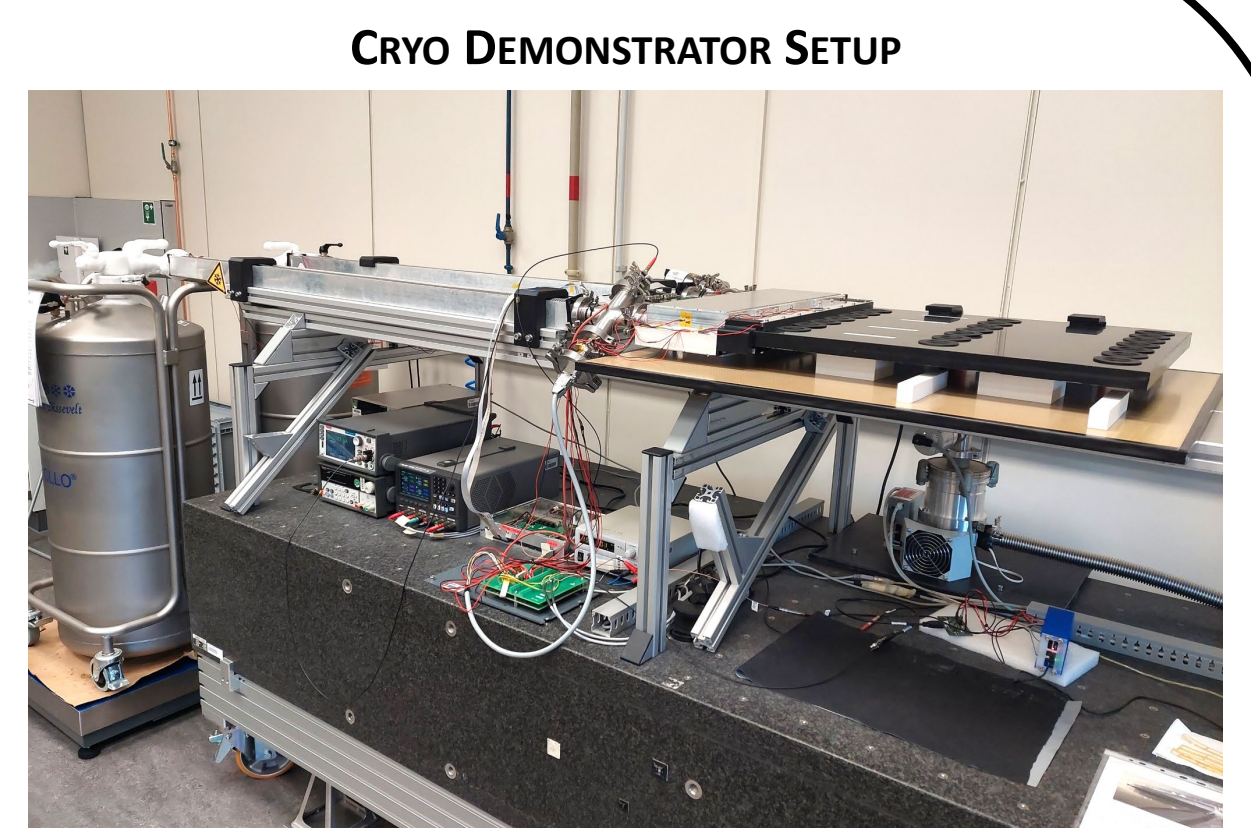
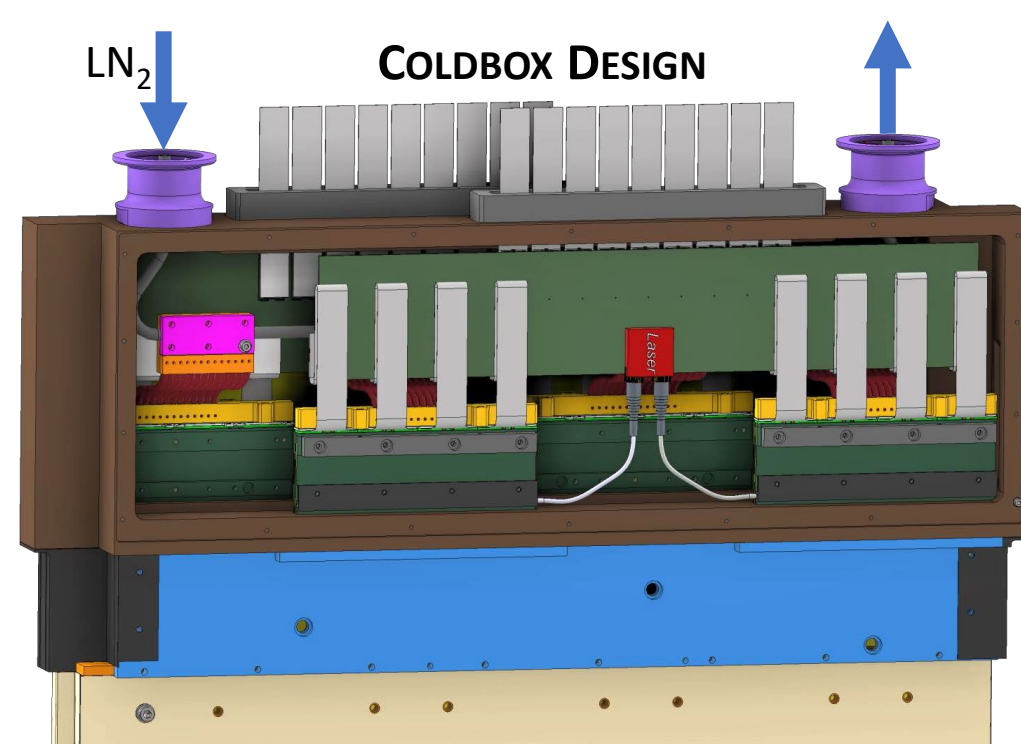
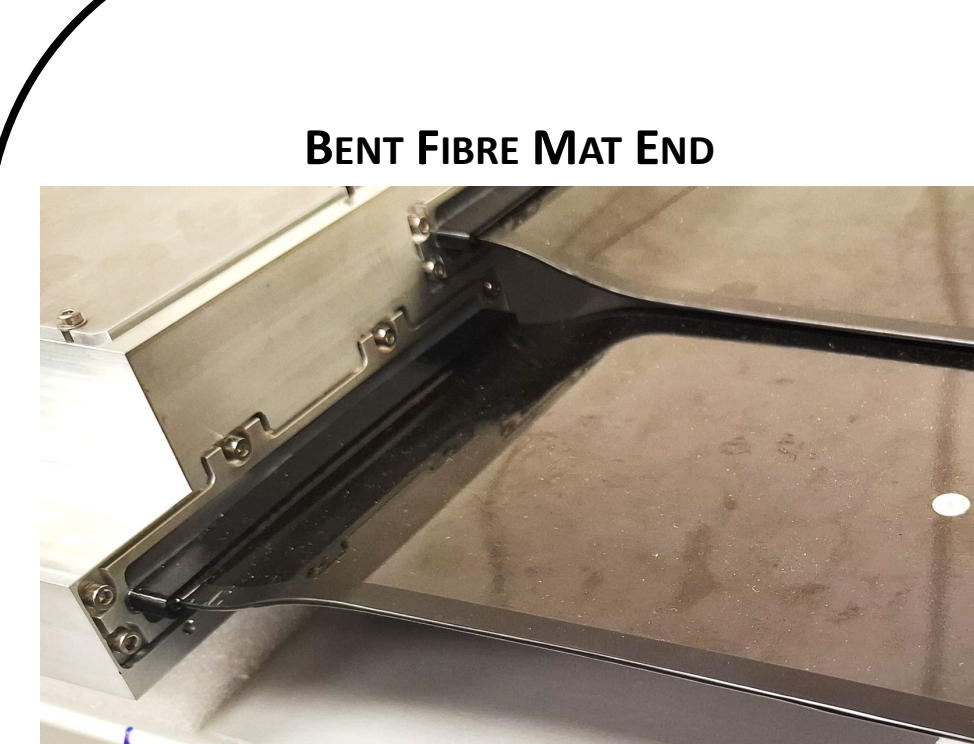
MAIN DESIGN FEATURES



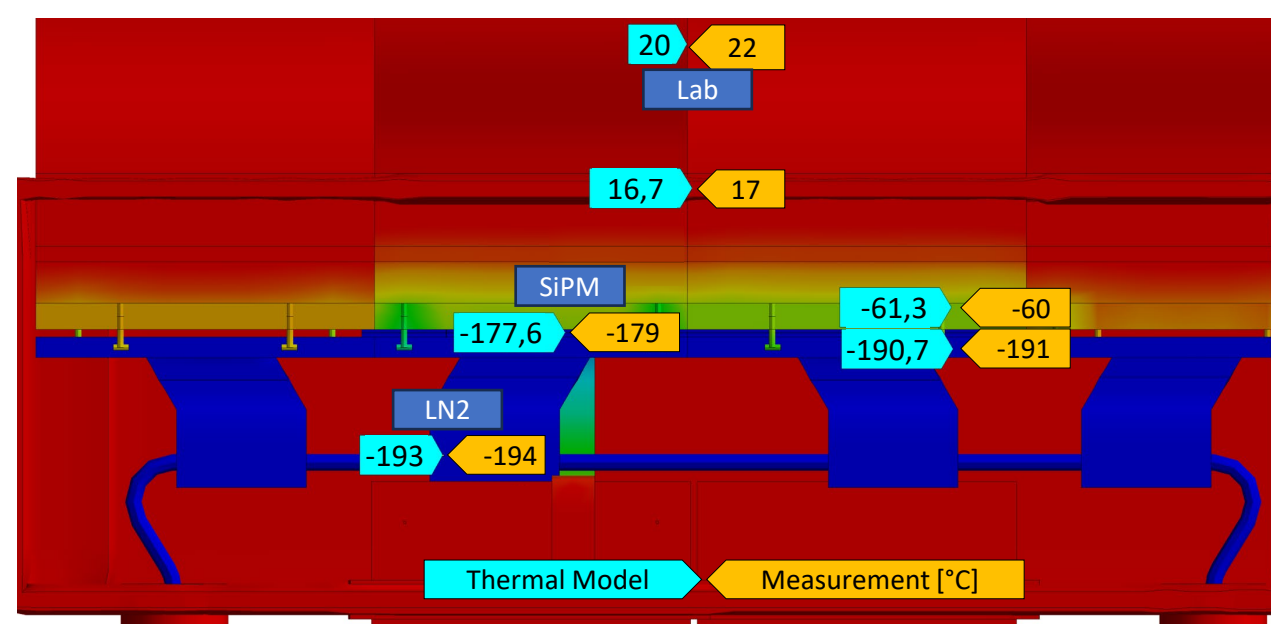
- Flexible fibre interface allows for:
 - Mechanical decoupling between sensitive feedthrough and fibre module
 - Two row arrangement of mats in the endpiece for space saving
 - Easy serial (linear) connection of coldboxes
 - Double row SiPM arrays for compact design and facilitation of alignment
- $T_{\text{SiPM}} < 90 \text{ K}$ (for LN_2 at 79 K) thanks to thermal spacers
- Measured heat load $9.6 \text{ W} \pm 20\%$ per box with 2 g/s LN_2 flow
- Pressure in the coldbox $\sim 2 \cdot 10^{-5} \text{ mbar}$ at cold
- Leak tightness tested with a pressure increase method $\sim 2 \cdot 10^{-5} \text{ mbar} \cdot \text{L/s}$



THERMALLY RESHAPED FIBRE MATS



THERMAL MODEL



MAIN DESIGN FEATURES

- Rigid mat thermally bent at 65°C to obtain the S-shape
- Light yield measurements show $\sim 1\%$ light loss compared to straight mat
- Thermal model calibrated to measurements
- SiPM at $92 \pm 3 \text{ K}$ (for LN_2 at 79 K)
- Measured and simulated heat load scaled to full coldbox $\sim 10 \text{ W}$
- LN_2 flow 2 g/s, pressure in the coldbox $\sim 1 \cdot 10^{-5} \text{ mbar}$
- Leak rate $6 \cdot 10^{-5} \text{ mbar} \cdot \text{L/s}$ with elastomer seals